

Liam Murphy

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EXPERIENCE

FSAE Aerodynamicist/Test Engineer

September 2024 - Present

CU Boulder Racing Team, Boulder, CO

- Designed team's first ground-effect driven undertray for 30% increased downforce with small drag penalty.
- Manufactured carbon composite aerodynamic elements and integrated them into the final vehicle assembly.
- Conducting suspension potentiometer testing to verify CFD acquired downforce numbers for the aero package. Will serve as a validation platform for any future aero package and chassis iterations.

Launch System Engineer

August 2026 - Present

CU Boulder SuperSpeed, Boulder, CO

- Design a bungee launch system for an electric ducted fan (EDF) drone targeting speeds above 200mph.
- Develop launcher to achieve an 18 m/s velocity at 8° launch angle while meeting portability requirements.

Surveillance Drone Engineer

April 2026 - Present

CU Boulder Robotics, Boulder, CO

- Designed small quadcopter with lidar array for mapping and relaying enemy robot positions in real time.
- The lidar footprint is designed to provide more than 60% field coverage from most hover locations.

Design Lead

August 2020 - June 2023

Vex Robotics Competition, Davie, FL

- Led design on mechanical solutions to game challenges.
- Finalist at Vex Worlds 2022 with team 4154X.

PROJECTS

FPV Racing/Cinematic Drones

September 2025 - Present

- Built racing micro-quadcopter with a target power to weight ratio (PWR) of 3+ focused on aerobatics.
- Designed 5" drone for cinematic flight shots using a custom 3d printed frame using carbon fiber filament.
- Iterative design process using flight/static test data to improve frame performances and avionics packaging.
- Selected avionics via a constraint driven trade study balancing weight, size, and performance requirements.

Quadcopter Motor Test Stand

August 2026 - Present

- Designed brushless DC motor test stand using a 5kg S-type load cell for axial thrust measurements to optimize drone frame designs and avionics selection.

EDUCATION

University of Colorado Boulder

May 2027 Graduation

Major: Aerospace Engineering

- Organizations: CU Boulder Racing Team FSAE Branch, CU Boulder SuperSpeed, CU Boulder Robotics

SKILLS

- **Technical:** Composites Manufacturing, Aerodynamic Design, Drone Design, CAD, CFD, MATLAB
- **Other:** Team Leadership, Project Management, Technical Communication